UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Dipartimento
di Fisica
e Astronomia
Galileo Galilei

No U-Turn Sampler

Adaptively Setting Path Lengths in Hamiltonian Monte Carlo

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August 23, 2026

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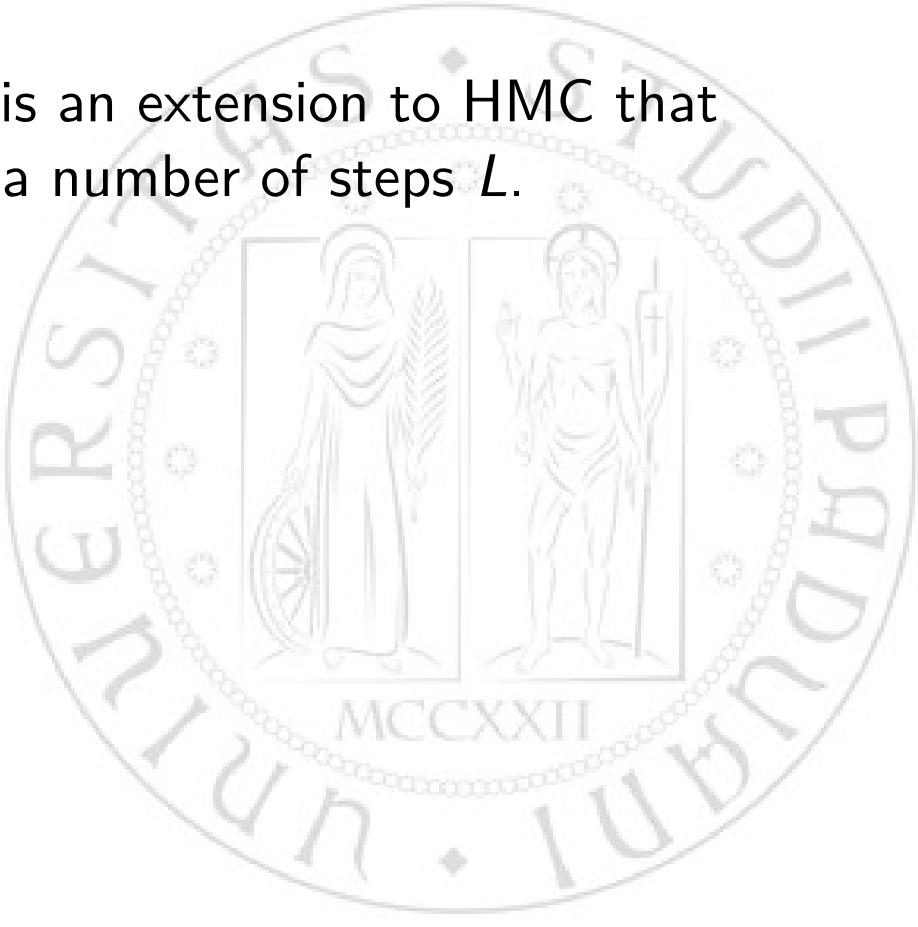

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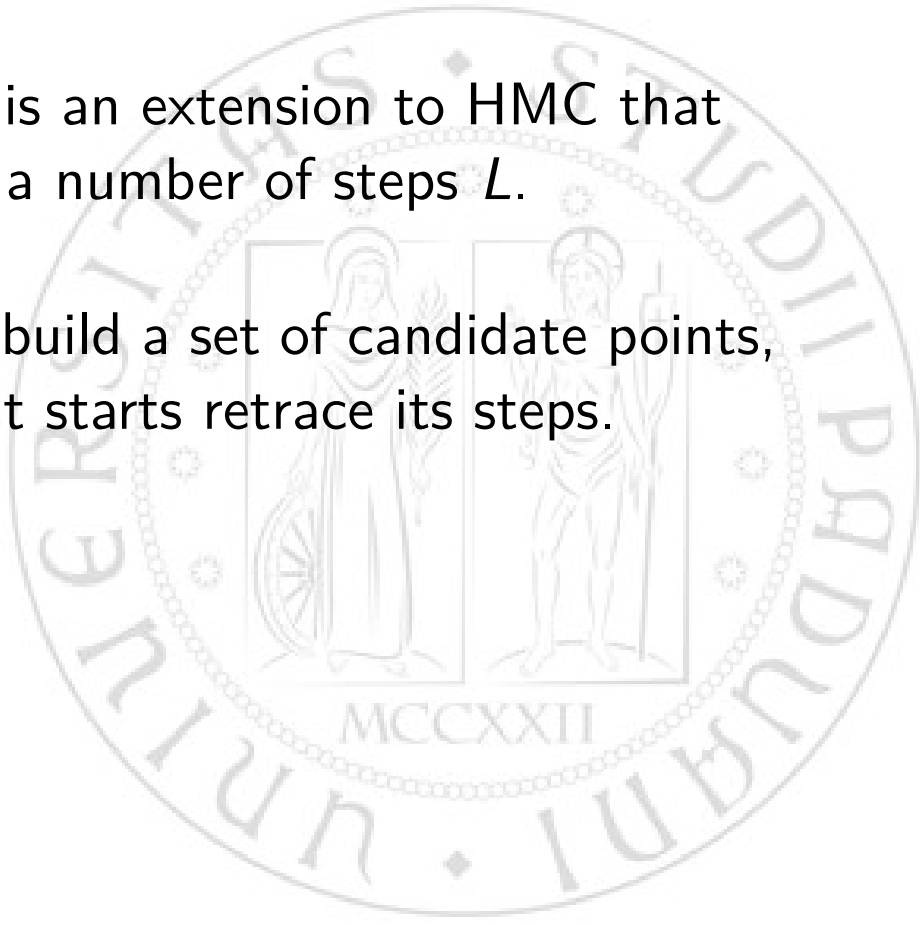

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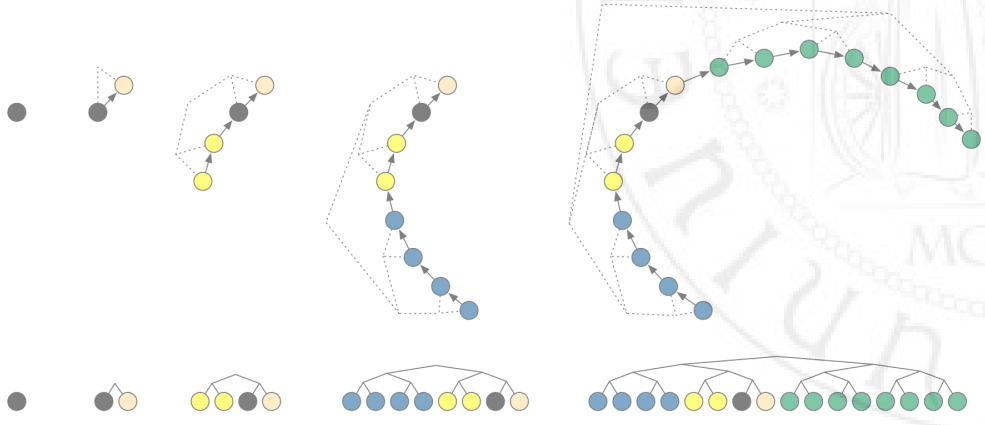

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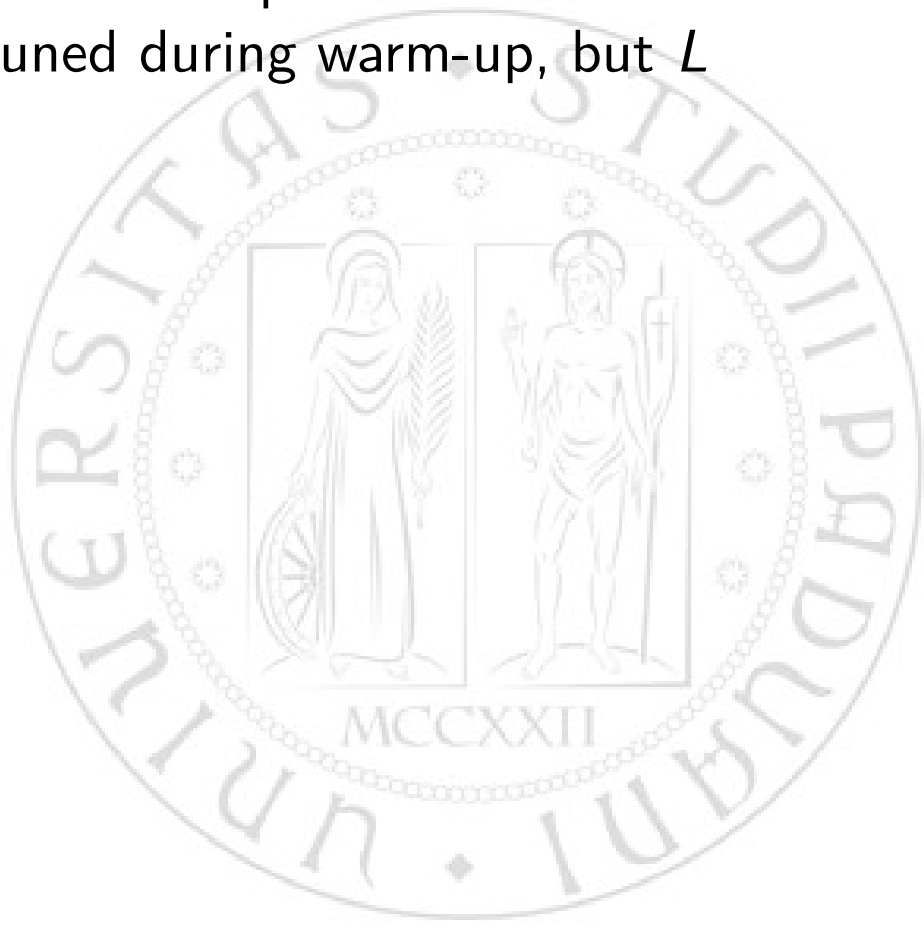

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Algorithm 1 Hamiltonian Monte Carlo

Given θ^0, ϵ, L, $\mathcal{L}$, M:
for $m = 1$ to M **do**
  Sample $r^0 \sim \mathcal{N}(0, I)$.
  Set $\theta^m \leftarrow \theta^{m-1}, \tilde{\theta} \leftarrow \theta^{m-1}, \tilde{r} \leftarrow r^0$.
  for $i = 1$ to L **do**
    Set $\tilde{\theta}, \tilde{r} \leftarrow \text{Leapfrog}(\tilde{\theta}, \tilde{r}, \epsilon)$.
  end for
  With probability $\alpha = \min\left\{1, \frac{\exp\{\mathcal{L}(\tilde{\theta}) - \frac{1}{2}\tilde{r}\cdot\tilde{r}\}}{\exp\{\mathcal{L}(\theta^{m-1}) - \frac{1}{2}r^0\cdot r^0\}}\right\}$, set $\theta^m \leftarrow \tilde{\theta}$, $r^m \leftarrow -\tilde{r}$.
end for

function Leapfrog(θ, r, ϵ)
Set $\tilde{r} \leftarrow r + (\epsilon/2)\nabla_\theta \mathcal{L}(\theta)$.
Set $\tilde{\theta} \leftarrow \theta + \epsilon\tilde{r}$.
Set $\tilde{r} \leftarrow \tilde{r} + (\epsilon/2)\nabla_\theta \mathcal{L}(\tilde{\theta})$.
return $\tilde{\theta}, \tilde{r}$.

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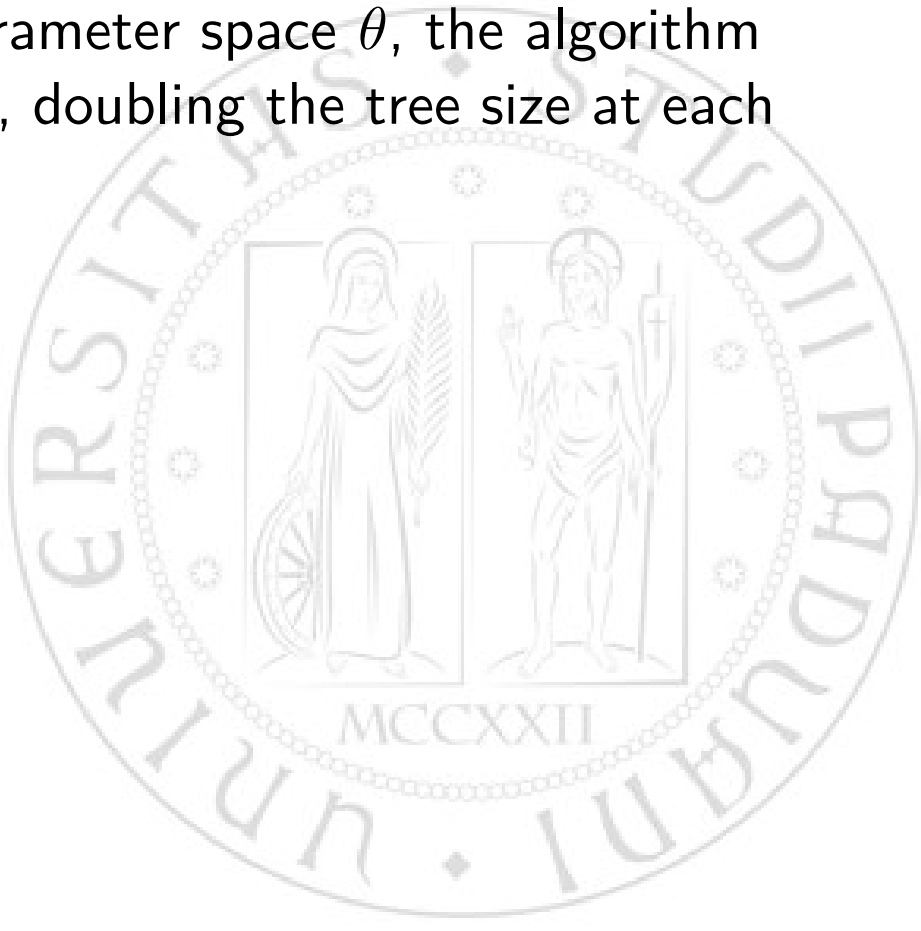

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- Starting from a point in the parameter space θ, the algorithm builds a set of candidate points, doubling the tree size at each iteration.
- To detect when to stop, the algorithm checks if the trajectory starts to turn back on itself: given a proposed point $\tilde{\theta}$, the algorithm checks if the dot product of the momentum vectors at θ and $\tilde{\theta}$ is negative.
  $(\theta - \tilde{\theta}) \cdot \frac{d}{dt}(\theta - \tilde{\theta}) = (\theta - \tilde{\theta}) \cdot \tilde{r} < 0$

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  $(\theta - \tilde{\theta}) \cdot \frac{d}{dt}(\theta - \tilde{\theta}) = (\theta - \tilde{\theta}) \cdot \tilde{r} < 0$
- To preserve time reversibility, NUTS runs the Hamiltonian dynamics in both forward and backward directions.

Probability slicing

To simplify the derivation and implementation of the algorithm, NUTS uses a slice variable u with conditional distribution

$$p(u|\theta, r) = \mathcal{U}(u; [0, exp\mathcal{L}(\theta) - \tfrac{1}{2} r \cdot r]).$$

This way $p(\theta, r|u) = \mathcal{U}(\theta, r; \theta', r'|exp\mathcal{L}(\theta) - \tfrac{1}{2} r \cdot r \geq u)$.

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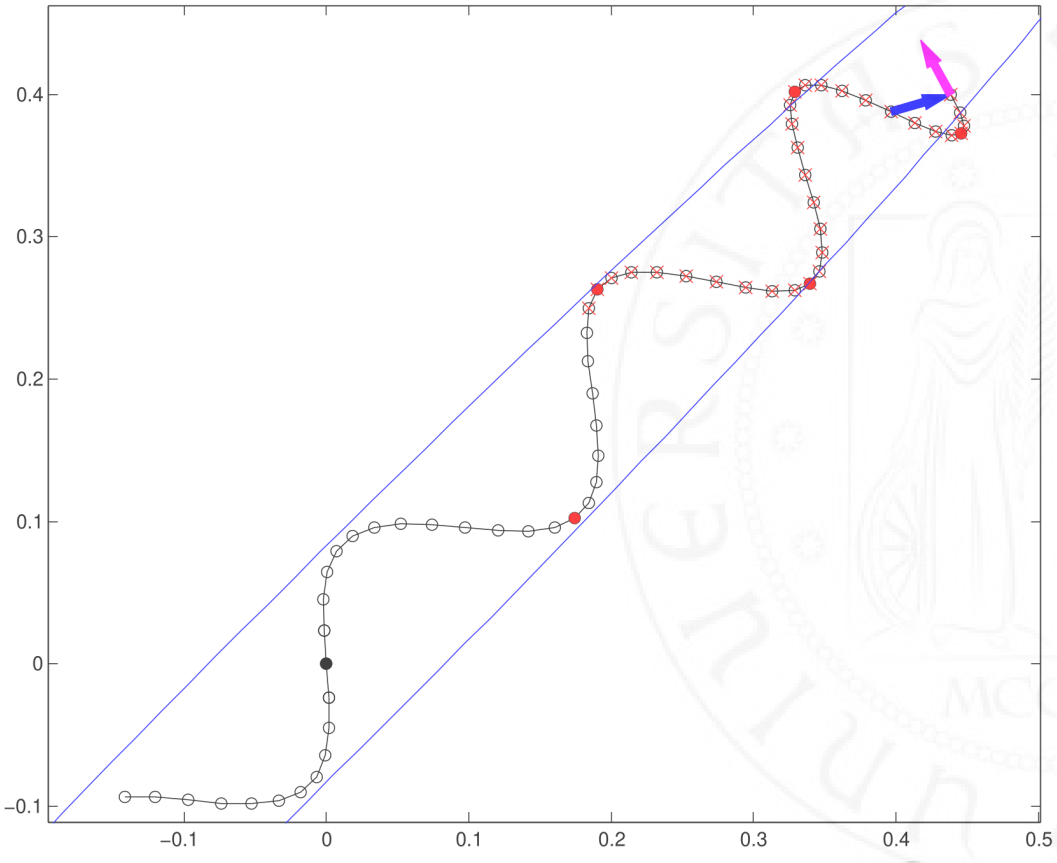
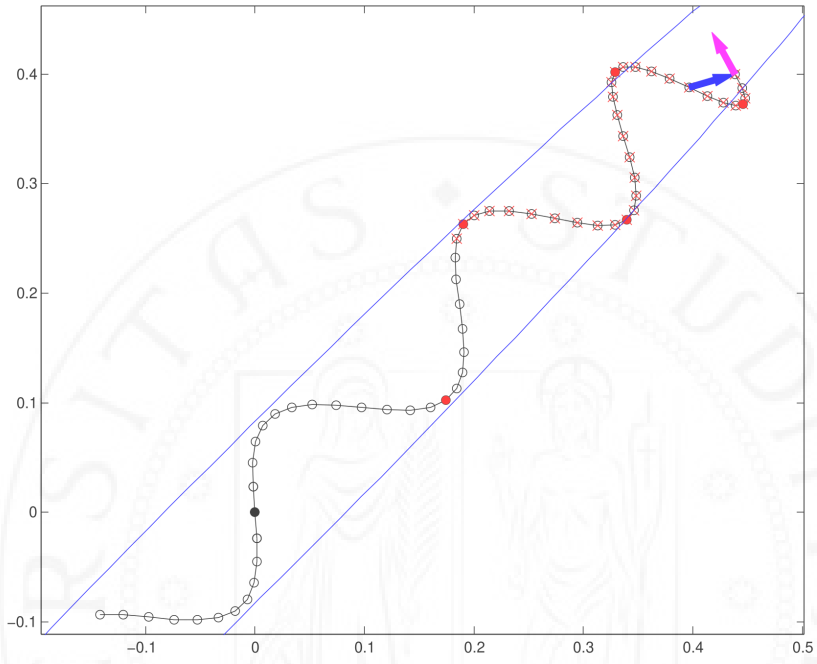


High-level NUTS algorithm

- The algorithm resamples $u|\theta, r$ at each iteration,



Example of a NUTS trajectory, with the slice variable u.

High-level NUTS algorithm

- The algorithm resamples $u|\theta, r$ at each iteration,
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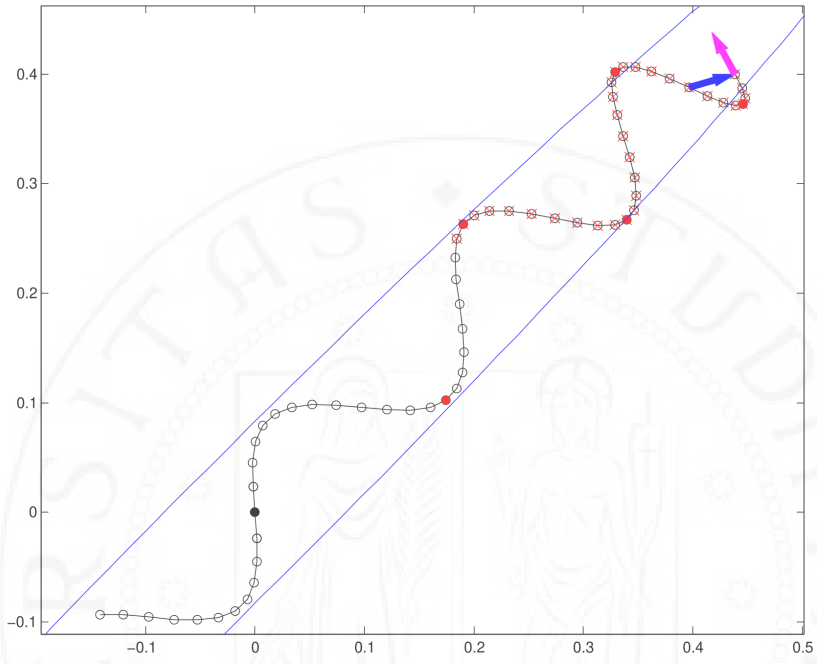

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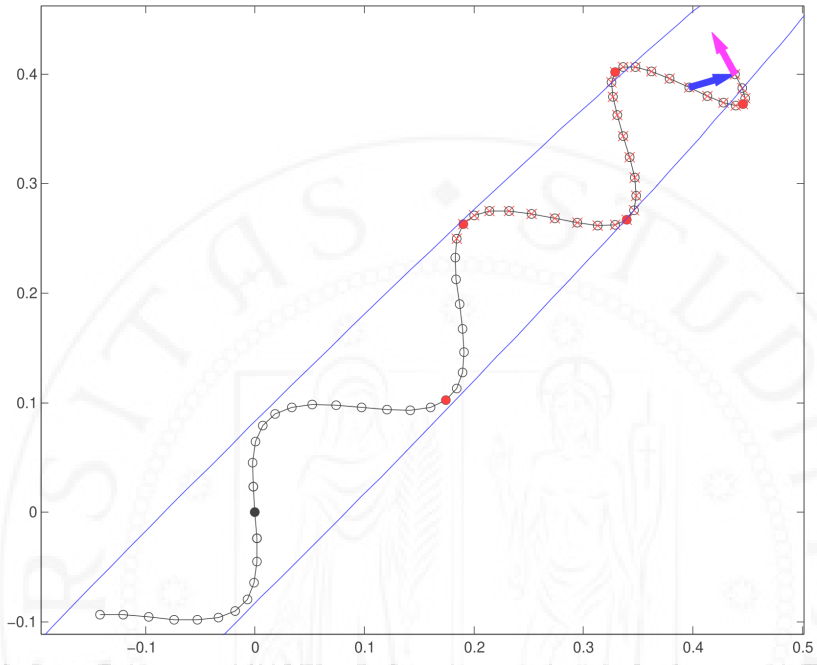

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- The algorithm resamples $u|\theta, r$ at each iteration,
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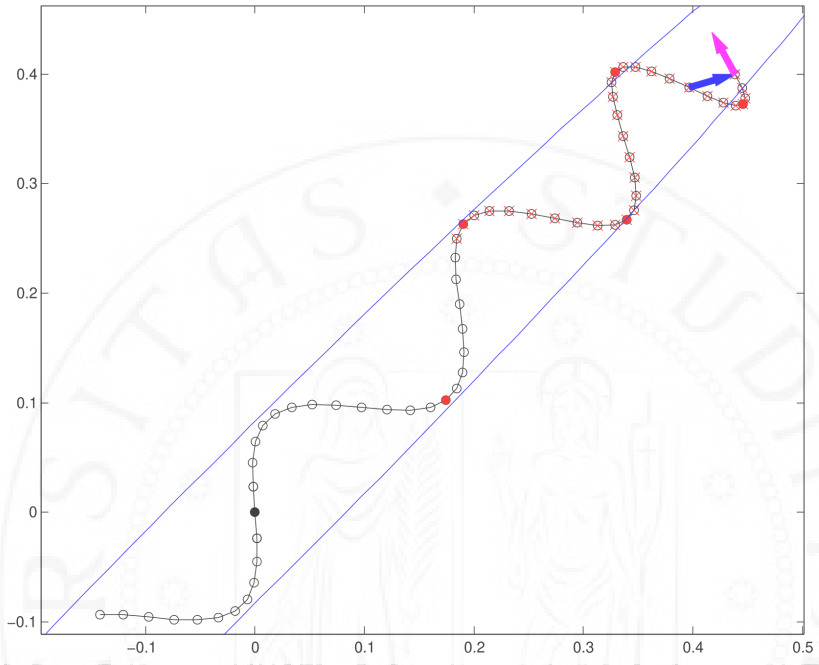
- This doubling process implicitly builds a balanced binary tree whose leaf nodes correspond to position-momentum states.



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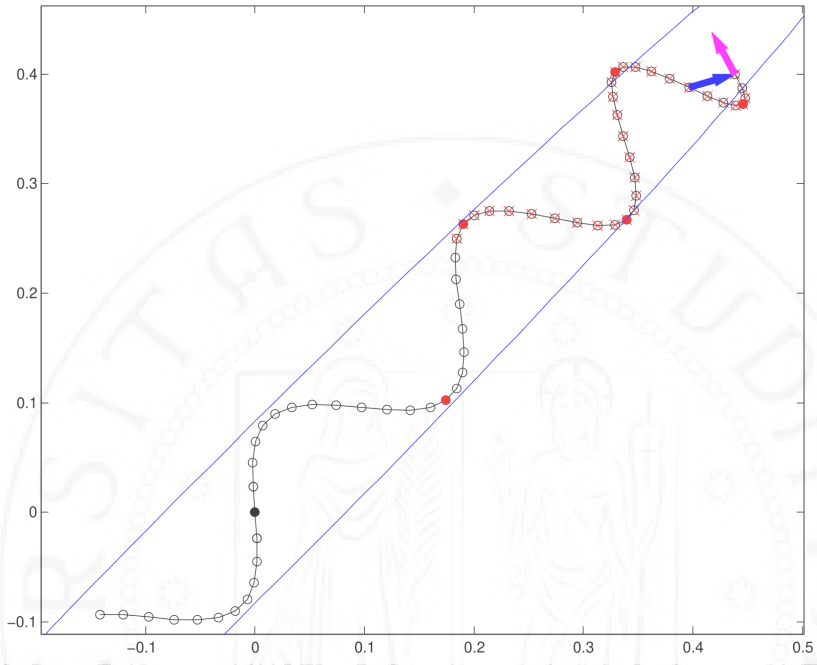
- The doubling is halted when the subtrajectory starts to double back on itself.



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Detailed NUTS algorithm

Given $\mathcal{B}$, set of position-momentum states traced by the leapfrog integrator, we pick deterministicly $\mathcal{C}$, the set of candidate points.

Algorithm 2 Naive No-U-Turn Sampler

Given θ^0, ϵ, $\mathcal{L}$, M:
for $m = 1$ to M **do**
  Resample $r^0 \sim \mathcal{N}(0, I)$.
  Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2} r^0 \cdot r^0\}])$
  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$.
    else
      $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$.
    end if
    if $s' = 1$ **then**
      $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$.
    end if
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
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  end while
  Sample θ^m, r uniformly at random from $\mathcal{C}$.
end for

function BuildTree$(\theta, r, u, v, j, \epsilon)$
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
  $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.
  $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$
  $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$.
  return $\theta', r', \theta', r', \mathcal{C}', s'$.
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- Transitions between states in $\mathcal{C}$ must preserve proability volume and satisfy detailed balance.
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- Each point in $\mathcal{C}$ must satisfy the slice variable condition $exp\mathcal{L}(\theta) - \frac{1}{2} r \cdot r \geq u$.
- Each point in $\mathcal{C}$ must have the same probability of constructing the trajectory $\mathcal{B}$ and candidate set $\mathcal{C}$.

Algorithm 2 Naive No-U-Turn Sampler

Given θ^0, ϵ, $\mathcal{L}$, M:
for $m = 1$ to M **do**
  Resample $r^0 \sim \mathcal{N}(0, I)$.
  Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2} r^0 \cdot r^0\}])$
  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$.
    else
      $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$.
    end if
    if $s' = 1$ **then**
      $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$.
    end if
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
    $j \leftarrow j + 1$.
  end while
  Sample θ^m, r uniformly at random from $\mathcal{C}$.
end for

function BuildTree$(\theta, r, u, v, j, \epsilon)$
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
  $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.
  $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$
  $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$.
  return $\theta', r', \theta', r', \mathcal{C}', s'$.
else
  Recursion—build the left and right subtrees.
  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon)$.
  if $v = -1$ **then**
    $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon)$.
  else
    $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon)$.
  end if
  $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
  $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$.
  return $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$.
end if

Detailed NUTS algorithm

We can ensure that with the following procedure at iteration t:

Algorithm 2 Naive No-U-Turn Sampler

Given θ^0, ϵ, $\mathcal{L}$, M:
for $m = 1$ to M **do**
  Resample $r^0 \sim \mathcal{N}(0, I)$.
  Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2} r^0 \cdot r^0\}])$
  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$.
    else
      $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$.
    end if
    if $s' = 1$ **then**
      $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$.
    end if
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
    $j \leftarrow j + 1$.
  end while
  Sample θ^m, r uniformly at random from $\mathcal{C}$.
end for

function BuildTree$(\theta, r, u, v, j, \epsilon)$
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
  $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.
  $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$
  $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$.
  return $\theta', r', \theta', r', \mathcal{C}', s'$.
else
  Recursion—build the left and right subtrees.
  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon)$.
  if $v = -1$ **then**
    $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon)$.
  else
    $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon)$.
  end if
  $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
  $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$.
  return $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$.
end if

Detailed NUTS algorithm

We can ensure that with the following procedure at iteration t:

- Sample $r \sim \mathcal{N}(0, I)$,

Algorithm 2 Naive No-U-Turn Sampler

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  Resample $r^0 \sim \mathcal{N}(0, I)$.
  Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2} r^0 \cdot r^0\}])$
  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$.
    else
      $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$.
    end if
    if $s' = 1$ **then**
      $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$.
    end if
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
    $j \leftarrow j + 1$.
  end while
  Sample θ^m, r uniformly at random from $\mathcal{C}$.
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function BuildTree$(\theta, r, u, v, j, \epsilon)$
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
  $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.
  $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$
  $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$.
  return $\theta', r', \theta', r', \mathcal{C}', s'$.
else
  Recursion—build the left and right subtrees.
  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon)$.
  if $v = -1$ **then**
    $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon)$.
  else
    $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon)$.
  end if
  $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
  $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$.
  return $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$.
end if

Detailed NUTS algorithm

We can ensure that with the following procedure at iteration t:

- Sample $r \sim \mathcal{N}(0, I)$,
- $u \sim \mathcal{U}(0, exp\mathcal{L}(\theta^t) - \frac{1}{2} r \cdot r)$.

Algorithm 2 Naive No-U-Turn Sampler

Given θ^0, ϵ, $\mathcal{L}$, M:
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  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$.
    else
      $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$.
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    if $s' = 1$ **then**
      $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$.
    end if
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
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function BuildTree$(\theta, r, u, v, j, \epsilon)$
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  $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$.
  return $\theta', r', \theta', r', \mathcal{C}', s'$.
else
  Recursion—build the left and right subtrees.
  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon)$.
  if $v = -1$ **then**
    $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon)$.
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  end if
  $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
  $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$.
  return $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$.
end if

Detailed NUTS algorithm

We can ensure that with the following procedure at iteration t:

- Sample $r \sim \mathcal{N}(0, I)$,
- $u \sim \mathcal{U}(0, exp\mathcal{L}(\theta^t) - \frac{1}{2} r \cdot r)$.
- Sample $\mathcal{B}, \mathcal{C} \sim p(\mathcal{B}, \mathcal{C} | \theta^t, r, u, \epsilon)$.

Algorithm 2 Naive No-U-Turn Sampler

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  Resample $r^0 \sim \mathcal{N}(0, I)$.
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  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$.
    else
      $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$.
    end if
    if $s' = 1$ **then**
      $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$.
    end if
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
    $j \leftarrow j + 1$.
  end while
  Sample θ^m, r uniformly at random from $\mathcal{C}$.
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function BuildTree$(\theta, r, u, v, j, \epsilon)$
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
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  $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$.
  return $\theta', r', \theta', r', \mathcal{C}', s'$.
else
  Recursion—build the left and right subtrees.
  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon)$.
  if $v = -1$ **then**
    $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon)$.
  else
    $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon)$.
  end if
  $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
  $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$.
  return $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$.
end if

Detailed NUTS algorithm

We can ensure that with the following procedure at iteration t:

- Sample $r \sim \mathcal{N}(0, I)$,
- $u \sim \mathcal{U}(0, exp\mathcal{L}(\theta^t) - \frac{1}{2} r \cdot r)$.
- Sample $\mathcal{B}, \mathcal{C} \sim p(\mathcal{B}, \mathcal{C} | \theta^t, r, u, \epsilon)$.
- Sample $\theta^{t+1}, r \sim T(\theta^t, r, \mathcal{C})$, transition kernel.

Algorithm 2 Naive No-U-Turn Sampler

Given θ^0, ϵ, $\mathcal{L}$, M:
for $m = 1$ to M **do**
  Resample $r^0 \sim \mathcal{N}(0, I)$.
  Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2} r^0 \cdot r^0\}])$
  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$.
    else
      $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$.
    end if
    if $s' = 1$ **then**
      $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$.
    end if
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
    $j \leftarrow j + 1$.
  end while
  Sample θ^m, r uniformly at random from $\mathcal{C}$.
end for

function BuildTree$(\theta, r, u, v, j, \epsilon)$
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
  $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.
  $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$
  $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$.
  return $\theta', r', \theta', r', \mathcal{C}', s'$.
else
  Recursion—build the left and right subtrees.
  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon)$.
  if $v = -1$ **then**
    $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon)$.
  else
    $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon)$.
  end if
  $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
  $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$.
  return $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$.
end if

Detailed NUTS algorithm

We can ensure that with the following procedure at iteration t:

- Sample $r \sim \mathcal{N}(0, I)$,
- $u \sim \mathcal{U}(0, exp\mathcal{L}(\theta^t) - \frac{1}{2} r \cdot r)$.
- Sample $\mathcal{B}, \mathcal{C} \sim p(\mathcal{B}, \mathcal{C} | \theta^t, r, u, \epsilon)$.
- Sample $\theta^{t+1}, r \sim T(\theta^t, r, \mathcal{C})$, transition kernel.

Where the transition kernel $T(\theta, r, \mathcal{C})$ has leave the distribution over $\mathcal{C}$ invariant, valid because ϵ is uniform on the elements of $\mathcal{C}$ and $p(\theta, r | u, \mathcal{B}, \mathcal{C}, \epsilon) \propto \mathbb{I}[(\theta, r) \in \mathcal{C}]$.

Algorithm 2 Naive No-U-Turn Sampler

Given θ^0, ϵ, $\mathcal{L}$, M:
for $m = 1$ to M **do**
  Resample $r^0 \sim \mathcal{N}(0, I)$.
  Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2} r^0 \cdot r^0\}])$
  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$.
    else
      $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$.
    end if
    if $s' = 1$ **then**
      $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$.
    end if
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
    $j \leftarrow j + 1$.
  end while
  Sample θ^m, r uniformly at random from $\mathcal{C}$.
end for

function BuildTree$(\theta, r, u, v, j, \epsilon)$
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
  $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.
  $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$
  $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$.
  return $\theta', r', \theta', r', \mathcal{C}', s'$.
else
  Recursion—build the left and right subtrees.
  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon)$.
  if $v = -1$ **then**
    $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon)$.
  else
    $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon)$.
  end if
  $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
  $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$.
  return $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$.
end if

Detailed NUTS algorithm

The trajectory $\mathcal{B}$ is built with a doubling binary tree structure:

Algorithm 2 Naive No-U-Turn Sampler

Given θ^0, ϵ, $\mathcal{L}$, M:
for $m = 1$ to M **do**
  Resample $r^0 \sim \mathcal{N}(0, I)$.
  Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2} r^0 \cdot r^0\}])$
  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$.
    else
      $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$.
    end if
    if $s' = 1$ **then**
      $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$.
    end if
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
    $j \leftarrow j + 1$.
  end while
  Sample θ^m, r uniformly at random from $\mathcal{C}$.
end for

function BuildTree$(\theta, r, u, v, j, \epsilon)$
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
  $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.
  $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$
  $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$.
  return $\theta', r', \theta', r', \mathcal{C}', s'$.
else
  Recursion—build the left and right subtrees.
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  if $v = -1$ **then**
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  end if
  $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
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Detailed NUTS algorithm

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  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.
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    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
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function BuildTree$(\theta, r, u, v, j, \epsilon)$
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  $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$.
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else
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  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon)$.
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Detailed NUTS algorithm

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The doubling process defines the distribution $p(\mathcal{B}|\theta^t, r, u, \epsilon)$.

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Detailed NUTS algorithm

- NUTS algorithm repeatedly calls the BuildTree function, returning left and right endpoints of the trajectory, and all visited states $\mathcal{C}'$.
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- NUTS algorithm repeatedly calls the BuildTree function, returning left and right endpoints of the trajectory, and all visited states $\mathcal{C}'$.
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- Otherwise, the new candidates $\mathcal{C}'$ are added to the set of candidate points $\mathcal{C}$.

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    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
    $j \leftarrow j + 1$.
  end while
  Sample θ^m, r uniformly at random from $\mathcal{C}$.
end for

function BuildTree$(\theta, r, u, v, j, \epsilon)$
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
  $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.
  $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$
  $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$.
  return $\theta', r', \theta', r', \mathcal{C}', s'$.
else
  Recursion—build the left and right subtrees.
  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon)$.
  if $v = -1$ **then**
    $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon)$.
  else
    $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon)$.
  end if
  $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
  $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$.
  return $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$.
end if

Detailed NUTS algorithm

- NUTS algorithm repeatedly calls the BuildTree function, returning left and right endpoints of the trajectory, and all visited states $\mathcal{C}'$.
- If one of the two stepping conditions is met, the algorithm terminates and the new candidates $\mathcal{C}'$ are rejected.
- Otherwise, the new candidates $\mathcal{C}'$ are added to the set of candidate points $\mathcal{C}$.
- If a global U turn condition is met, the doubling process is terminated.

Algorithm 2 Naive No-U-Turn Sampler

Given θ^0, ϵ, $\mathcal{L}$, M:
for $m = 1$ to M **do**
  Resample $r^0 \sim \mathcal{N}(0, I)$.
  Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2} r^0 \cdot r^0\}])$
  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$.
    else
      $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$.
    end if
    if $s' = 1$ **then**
      $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$.
    end if
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
    $j \leftarrow j + 1$.
  end while
  Sample θ^m, r uniformly at random from $\mathcal{C}$.
end for

function BuildTree$(\theta, r, u, v, j, \epsilon)$
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
  $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.
  $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$
  $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$.
  return $\theta', r', \theta', r', \mathcal{C}', s'$.
else
  Recursion—build the left and right subtrees.
  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon)$.
  if $v = -1$ **then**
    $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon)$.
  else
    $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon)$.
  end if
  $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
  $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$.
  return $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$.
end if

Detailed NUTS algorithm

- NUTS algorithm repeatedly calls the BuildTree function, returning left and right endpoints of the trajectory, and all visited states $\mathcal{C}'$.
- If one of the two stepping conditions is met, the algorithm terminates and the new candidates $\mathcal{C}'$ are rejected.
- Otherwise, the new candidates $\mathcal{C}'$ are added to the set of candidate points $\mathcal{C}$.
- If a global U turn condition is met, the doubling process is terminated.
- The new position-momentum state (θ^{t+1}, r) is sampled from the set of candidate points $\mathcal{C}$.

Algorithm 2 Naive No-U-Turn Sampler

Given θ^0, ϵ, $\mathcal{L}$, M:
for $m = 1$ to M **do**
  Resample $r^0 \sim \mathcal{N}(0, I)$.
  Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2} r^0 \cdot r^0\}])$
  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0$, $r^+ = r^0$, $j = 0$, $\mathcal{C} = \{(\theta^{m-1}, r^0)\}$, $s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, -, -, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon)$.
    else
      $-, -, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon)$.
    end if
    if $s' = 1$ **then**
      $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}'$.
    end if
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
    $j \leftarrow j + 1$.
  end while
  Sample θ^m, r uniformly at random from $\mathcal{C}$.
end for

function BuildTree$(\theta, r, u, v, j, \epsilon)$
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
  $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.
  $\mathcal{C}' \leftarrow \begin{cases} \{(\theta', r')\} & \text{if } u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\} \\ \emptyset & \text{else} \end{cases}$
  $s' \leftarrow \mathbb{I}[\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' > \log u - \Delta_{\max}]$.
  return $\theta', r', \theta', r', \mathcal{C}', s'$.
else
  Recursion—build the left and right subtrees.
  $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s' \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon)$.
  if $v = -1$ **then**
    $\theta^-, r^-, -, -, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon)$.
  else
    $-, -, \theta^+, r^+, \mathcal{C}'', s'' \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon)$.
  end if
  $s' \leftarrow s' s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
  $\mathcal{C}' \leftarrow \mathcal{C}' \cup \mathcal{C}''$.
  return $\theta^-, r^-, \theta^+, r^+, \mathcal{C}', s'$.
end if

Optimized NUTS algorithm

Some optimizations can be made to the NUTS algorithm:

Algorithm 6 No-U-Turn Sampler with Dual Averaging

Given θ^0, δ, $\mathcal{L}$, M, M^{adapt}:
Set $\epsilon_0 = \text{FindReasonableEpsilon}(\theta)$, $\mu = \log(10\epsilon_0)$, $\bar{\epsilon}_0 = 1$, $\bar{H}_0 = 0$, $\gamma = 0.05$, $t_0 = 10$, $\kappa = 0.75$.
for $m = 1$ to M **do**
  Sample $r^0 \sim \mathcal{N}(0, I)$.
  Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2} r^0 \cdot r^0\}])$
  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0, r^+ = r^0$, $j = 0, \theta^m = \theta^{m-1}, n = 1, s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, -, -, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon_{m-1}\theta^{m-1}, r^0)$.
    else
      $-, -, \theta^+, r^+, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon_{m-1}, \theta^{m-1}, r^0)$.
    end if
    if $s' = 1$ **then**
      With probability $\min\{1, \frac{n'}{n}\}$, set $\theta^m \leftarrow \theta'$.
    end if
    $n \leftarrow n + n'$.
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
    $j \leftarrow j + 1$.
  end while
  if $m \leq M^{\text{adapt}}$ **then**
    Set $\bar{H}_m = \left(1 - \frac{1}{m + t_0}\right) \bar{H}_{m-1} + \frac{1}{m + t_0}(\delta - \frac{\alpha}{n_\alpha})$.
    Set $\log \epsilon_m = \mu - \frac{\sqrt{m}}{\gamma} \bar{H}_m$, $\log \bar{\epsilon}_m = m^{-\kappa} \log \epsilon_m + (1 - m^{-\kappa}) \log \bar{\epsilon}_{m-1}$.
  else
    Set $\epsilon_m = \bar{\epsilon}_{M^{\text{adapt}}}$.
  end if
end for

function $\text{BuildTree}(\theta, r, u, v, j, \epsilon, \theta^0, r^0)$
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
  $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.
  $n' \leftarrow \mathbb{I}[u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\}]$.
  $s' \leftarrow \mathbb{I}[u < \exp\{\Delta_{\max} + \mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\}]$.
  return $\theta', r', \theta', r', \theta', n', s', \min\{1, \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' - \mathcal{L}(\theta^0) + \frac{1}{2} r^0 \cdot r^0\}\}, 1$.
else
  Recursion—implicitly build the left and right subtrees.
  $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon, \theta^0, r^0)$.
  if $s' = 1$ **then**
    if $v = -1$ **then**
      $\theta^-, r^-, -, -, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon, \theta^0, r^0)$.
    else
      $-, -, \theta^+, r^+, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon, \theta^0, r^0)$.
    end if
    With probability $\frac{n''}{n' + n''}$, set $\theta' \leftarrow \theta''$.
    Set $\alpha' \leftarrow \alpha' + \alpha''$, $n'_\alpha \leftarrow n'_\alpha + n''_\alpha$.
    $s' \leftarrow s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$
    $n' \leftarrow n' + n''$
  end if
  return $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha$.
end if

Optimized NUTS algorithm

Some optimizations can be made to the NUTS algorithm:

- Early stopping when a stopping criterion is met mid-process.

Algorithm 6 No-U-Turn Sampler with Dual Averaging

Given θ^0, δ, $\mathcal{L}$, M, M^{adapt}:
Set $\epsilon_0 = \text{FindReasonableEpsilon}(\theta)$, $\mu = \log(10\epsilon_0)$, $\bar{\epsilon}_0 = 1$, $\bar{H}_0 = 0$, $\gamma = 0.05$, $t_0 = 10$, $\kappa = 0.75$.
for $m = 1$ to M **do**
  Sample $r^0 \sim \mathcal{N}(0, I)$.
  Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2} r^0 \cdot r^0\}])$
  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0, r^+ = r^0$, $j = 0, \theta^m = \theta^{m-1}, n = 1, s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, -, -, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon_{m-1}\theta^{m-1}, r^0)$.
    else
      $-, -, \theta^+, r^+, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon_{m-1}, \theta^{m-1}, r^0)$.
    end if
    if $s' = 1$ **then**
      With probability $\min\{1, \frac{n'}{n}\}$, set $\theta^m \leftarrow \theta'$.
    end if
    $n \leftarrow n + n'$.
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
    $j \leftarrow j + 1$.
  end while
  if $m \leq M^{\text{adapt}}$ **then**
    Set $\bar{H}_m = \left(1 - \frac{1}{m + t_0}\right) \bar{H}_{m-1} + \frac{1}{m + t_0}(\delta - \frac{\alpha}{n_\alpha})$.
    Set $\log \epsilon_m = \mu - \frac{\sqrt{m}}{\gamma} \bar{H}_m$, $\log \bar{\epsilon}_m = m^{-\kappa} \log \epsilon_m + (1 - m^{-\kappa}) \log \bar{\epsilon}_{m-1}$.
  else
    Set $\epsilon_m = \bar{\epsilon}_{M^{\text{adapt}}}$.
  end if
end for

function $\text{BuildTree}(\theta, r, u, v, j, \epsilon, \theta^0, r^0)$
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
  $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.
  $n' \leftarrow \mathbb{I}[u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\}]$.
  $s' \leftarrow \mathbb{I}[u < \exp\{\Delta_{\max} + \mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\}]$.
  return $\theta', r', \theta', r', \theta', n', s', \min\{1, \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' - \mathcal{L}(\theta^0) + \frac{1}{2} r^0 \cdot r^0\}\}, 1$.
else
  Recursion—implicitly build the left and right subtrees.
  $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon, \theta^0, r^0)$.
  if $s' = 1$ **then**
    if $v = -1$ **then**
      $\theta^-, r^-, -, -, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon, \theta^0, r^0)$.
    else
      $-, -, \theta^+, r^+, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon, \theta^0, r^0)$.
    end if
    With probability $\frac{n''}{n' + n''}$, set $\theta' \leftarrow \theta''$.
    Set $\alpha' \leftarrow \alpha' + \alpha''$, $n'_\alpha \leftarrow n'_\alpha + n''_\alpha$.
    $s' \leftarrow s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$
    $n' \leftarrow n' + n''$
  end if
  return $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha$.
end if

Optimized NUTS algorithm

Some optimizations can be made to the NUTS algorithm:

- Early stopping when a stopping criterion is met mid-process.
- Single Proposed State kept in memory, by sampling a candidate for each subtree and drawing from them using their relative weights (i.e. the size of the subtree).

Algorithm 6 No-U-Turn Sampler with Dual Averaging

Given θ^0, δ, $\mathcal{L}$, M, M^{adapt}:
Set $\epsilon_0 = \text{FindReasonableEpsilon}(\theta)$, $\mu = \log(10\epsilon_0)$, $\bar{\epsilon}_0 = 1$, $\bar{H}_0 = 0$, $\gamma = 0.05$, $t_0 = 10$, $\kappa = 0.75$.
for $m = 1$ to M **do**
  Sample $r^0 \sim \mathcal{N}(0, I)$.
  Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2} r^0 \cdot r^0\}])$
  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0, r^+ = r^0$, $j = 0, \theta^m = \theta^{m-1}, n = 1, s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, -, -, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon_{m-1} \theta^{m-1}, r^0)$.
    else
      $-, -, \theta^+, r^+, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon_{m-1}, \theta^{m-1}, r^0)$.
    end if
    if $s' = 1$ **then**
      With probability $\min\{1, \frac{n'}{n}\}$, set $\theta^m \leftarrow \theta'$.
    end if
    $n \leftarrow n + n'$.
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
    $j \leftarrow j + 1$.
  end while
  if $m \leq M^{\text{adapt}}$ **then**
    Set $\bar{H}_m = \left(1 - \frac{1}{m + t_0}\right) \bar{H}_{m-1} + \frac{1}{m + t_0} (\delta - \frac{\alpha}{n_\alpha})$.
    Set $\log \epsilon_m = \mu - \frac{\sqrt{m}}{\gamma} \bar{H}_m$, $\log \bar{\epsilon}_m = m^{-\kappa} \log \epsilon_m + (1 - m^{-\kappa}) \log \bar{\epsilon}_{m-1}$.
  else
    Set $\epsilon_m = \bar{\epsilon}_{M^{\text{adapt}}}$.
  end if
end for

function $\text{BuildTree}(\theta, r, u, v, j, \epsilon, \theta^0, r^0)$
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
  $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.
  $n' \leftarrow \mathbb{I}[u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\}]$.
  $s' \leftarrow \mathbb{I}[u < \exp\{\Delta_{\max} + \mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\}]$.
  return $\theta', r', \theta', r', \theta', n', s', \min\{1, \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' - \mathcal{L}(\theta^0) + \frac{1}{2} r^0 \cdot r^0\}\}, 1$.
else
  Recursion—implicitly build the left and right subtrees.
  $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon, \theta^0, r^0)$.
  if $s' = 1$ **then**
    if $v = -1$ **then**
      $\theta^-, r^-, -, -, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon, \theta^0, r^0)$.
    else
      $-, -, \theta^+, r^+, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon, \theta^0, r^0)$.
    end if
    With probability $\frac{n''}{n' + n''}$, set $\theta' \leftarrow \theta''$.
    Set $\alpha' \leftarrow \alpha' + \alpha''$, $n'_\alpha \leftarrow n'_\alpha + n''_\alpha$.
    $s' \leftarrow s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$
    $n' \leftarrow n' + n''$
  end if
  return $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha$.
end if

Optimized NUTS algorithm

Some optimizations can be made to the NUTS algorithm:

- Early stopping when a stopping criterion is met mid-process.
- Single Proposed State kept in memory, by sampling a candidate for each subtree and drawing from them using their relative weights (i.e. the size of the subtree).
- Adaptive tuning of the step size parameter ϵ, using dual averaging.

Algorithm 6 No-U-Turn Sampler with Dual Averaging

Given θ^0, δ, $\mathcal{L}$, M, M^{adapt}:
Set $\epsilon_0 = \text{FindReasonableEpsilon}(\theta)$, $\mu = \log(10\epsilon_0)$, $\bar{\epsilon}_0 = 1$, $\bar{H}_0 = 0$, $\gamma = 0.05$, $t_0 = 10$, $\kappa = 0.75$.
for $m = 1$ to M **do**
  Sample $r^0 \sim \mathcal{N}(0, I)$.
  Resample $u \sim \text{Uniform}([0, \exp\{\mathcal{L}(\theta^{m-1} - \frac{1}{2} r^0 \cdot r^0\}])$
  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0, r^+ = r^0$, $j = 0, \theta^m = \theta^{m-1}, n = 1, s = 1$.
  while $s = 1$ **do**
    Choose a direction $v_j \sim \text{Uniform}(\{-1, 1\})$.
    if $v_j = -1$ **then**
      $\theta^-, r^-, -, -, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v_j, j, \epsilon_{m-1} \theta^{m-1}, r^0)$.
    else
      $-, -, \theta^+, r^+, \theta', n', s', \alpha, n_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v_j, j, \epsilon_{m-1}, \theta^{m-1}, r^0)$.
    end if
    if $s' = 1$ **then**
      With probability $\min\{1, \frac{n'}{n}\}$, set $\theta^m \leftarrow \theta'$.
    end if
    $n \leftarrow n + n'$.
    $s \leftarrow s' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$.
    $j \leftarrow j + 1$.
  end while
  if $m \leq M^{\text{adapt}}$ **then**
    Set $\bar{H}_m = \left(1 - \frac{1}{m + t_0}\right) \bar{H}_{m-1} + \frac{1}{m + t_0} (\delta - \frac{\alpha}{n_\alpha})$.
    Set $\log \epsilon_m = \mu - \frac{\sqrt{m}}{\gamma} \bar{H}_m$, $\log \bar{\epsilon}_m = m^{-\kappa} \log \epsilon_m + (1 - m^{-\kappa}) \log \bar{\epsilon}_{m-1}$.
  else
    Set $\epsilon_m = \bar{\epsilon}_{M^{\text{adapt}}}$.
  end if
end for

function $\text{BuildTree}(\theta, r, u, v, j, \epsilon, \theta^0, r^0)$
if $j = 0$ **then**
  Base case—take one leapfrog step in the direction v.
  $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, v\epsilon)$.
  $n' \leftarrow \mathbb{I}[u \leq \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\}]$.
  $s' \leftarrow \mathbb{I}[u < \exp\{\Delta_{\max} + \mathcal{L}(\theta') - \frac{1}{2} r' \cdot r'\}]$.
  return $\theta', r', \theta', r', \theta', n', s', \min\{1, \exp\{\mathcal{L}(\theta') - \frac{1}{2} r' \cdot r' - \mathcal{L}(\theta^0) + \frac{1}{2} r^0 \cdot r^0\}\}, 1$.
else
  Recursion—implicitly build the left and right subtrees.
  $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha \leftarrow \text{BuildTree}(\theta, r, u, v, j - 1, \epsilon, \theta^0, r^0)$.
  if $s' = 1$ **then**
    if $v = -1$ **then**
      $\theta^-, r^-, -, -, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^-, r^-, u, v, j - 1, \epsilon, \theta^0, r^0)$.
    else
      $-, -, \theta^+, r^+, \theta'', n'', s'', \alpha'', n''_\alpha \leftarrow \text{BuildTree}(\theta^+, r^+, u, v, j - 1, \epsilon, \theta^0, r^0)$.
    end if
    With probability $\frac{n''}{n' + n''}$, set $\theta' \leftarrow \theta''$.
    Set $\alpha' \leftarrow \alpha' + \alpha''$, $n'_\alpha \leftarrow n'_\alpha + n''_\alpha$.
    $s' \leftarrow s'' \mathbb{I}[(\theta^+ - \theta^-) \cdot r^- \geq 0] \mathbb{I}[(\theta^+ - \theta^-) \cdot r^+ \geq 0]$
    $n' \leftarrow n' + n''$
  end if
  return $\theta^-, r^-, \theta^+, r^+, \theta', n', s', \alpha', n'_\alpha$.
end if

Dual Averaging for ϵ

ϵ is set using stochastic optimization with vanishing adaptation.

Algorithm 6 No-U-Turn Sampler with Dual Averaging

Given θ^0, δ, $\mathcal{L}$, M, M^{adapt}:
Set $\epsilon_0 = \text{FindReasonableEpsilon}(\theta), \mu = \log(10\epsilon_0), \bar{\epsilon}_0 = 1, \bar{H}_0 = 0, \gamma = 0.05, t_0 = 10, \kappa = 0.75$.
for $m = 1$ to M **do**
  Sample $r^0 \sim \mathcal{N}(0, I)$.
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  Initialize $\theta^- = \theta^{m-1}$, $\theta^+ = \theta^{m-1}$, $r^- = r^0, r^+ = r^0, j = 0, \theta^m = \theta^{m-1}, n = 1, s = 1$.
  while $s = 1$ **do**
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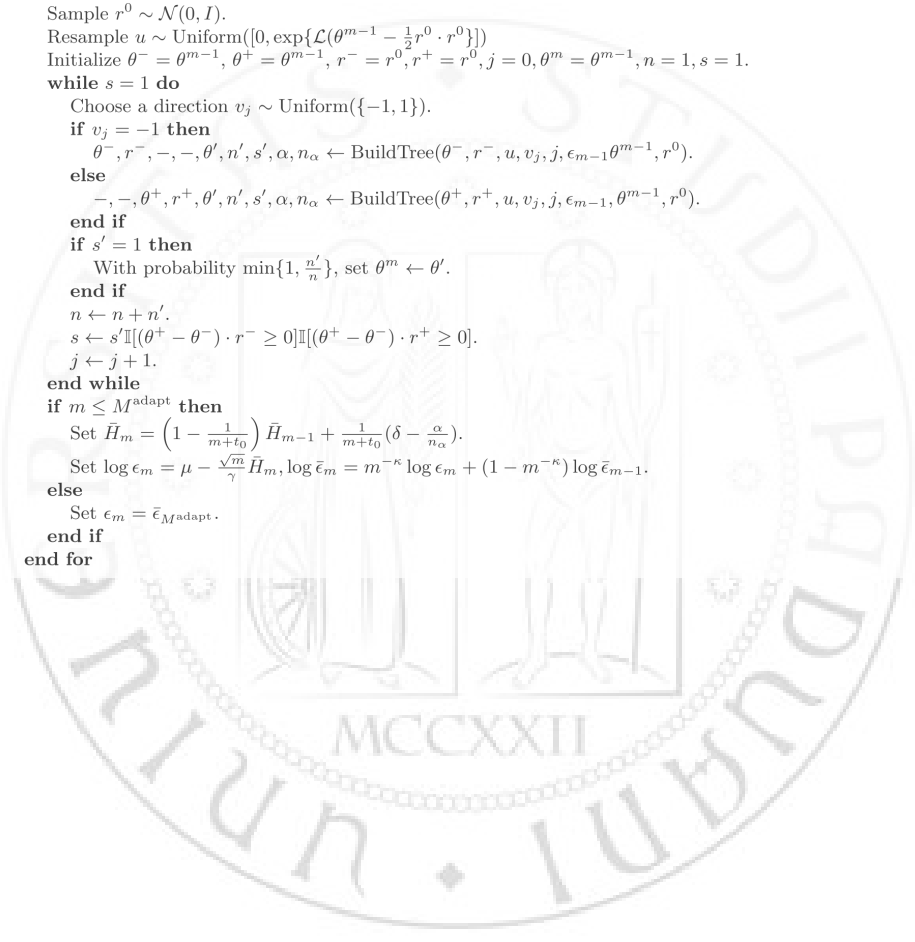

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    Set $\epsilon_m = \bar{\epsilon}_{M^{\text{adapt}}}$.
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$\mu = \log 10\epsilon_1$ is a freely chosen point, $\gamma = 0.05$ is a free parameter for the speed of convergence, and $t_0 = 10$ is a free parameter that stabilizes the initial iterations and $k = 0.75 \in (0.5, 1]$.

Heuristic function for ϵ

Convergence of Dual Averaging for ϵ is faster starting from a reasonable setting of parameters.

Algorithm 4 Heuristic for choosing an initial value of ϵ

function FindReasonableEpsilon(θ)
Initialize $\epsilon = 1$, $r \sim \mathcal{N}(0, I)$. (I denotes the identity matrix.)
Set $\theta', r' \leftarrow \text{Leapfrog}(\theta, r, \epsilon)$.
$a \leftarrow 2\mathbb{I}\left[\frac{p(\theta', r')}{p(\theta, r)} > 0.5\right] - 1$.
while $\left(\frac{p(\theta', r')}{p(\theta, r)}\right)^a > 2^{-a}$ **do**
  $\epsilon \leftarrow 2^a \epsilon$.
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- This heuristic doubles or halves the value of ϵ_1 until the acceptance probability of the Langevin (Leapfrog) proposal crosses 0.5.

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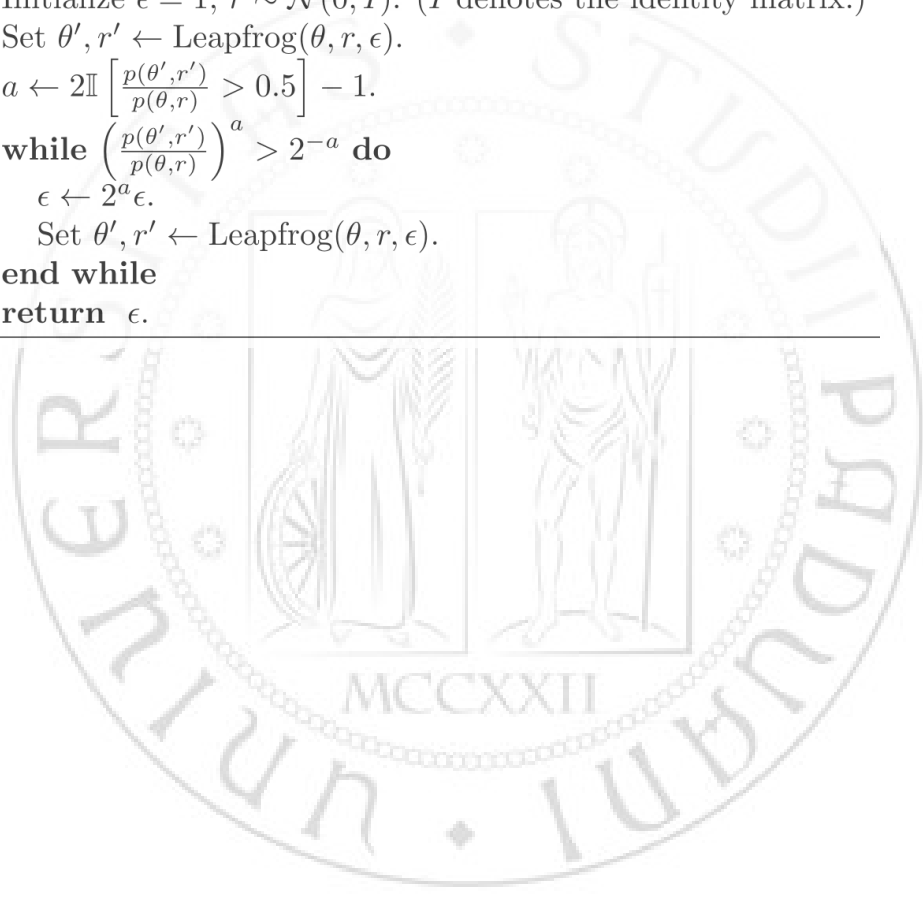

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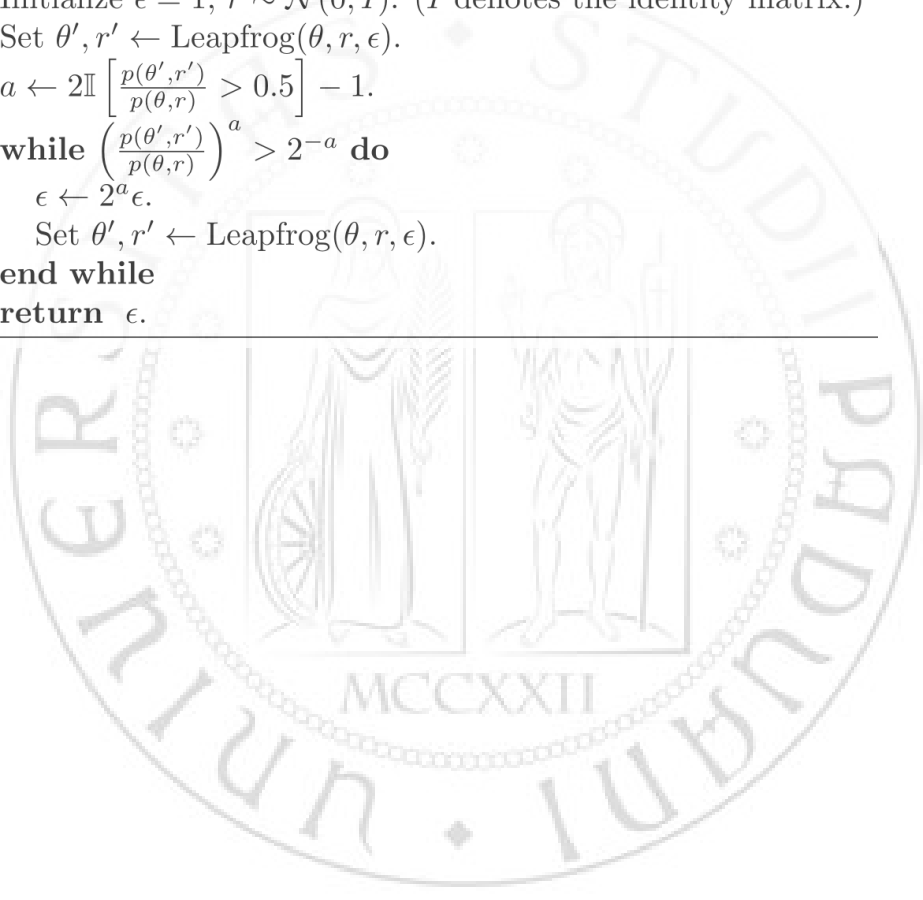

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$$H^{NUTS} = \frac{1}{|\mathcal{B}_t^f|} \sum_{\theta, r \in \mathcal{B}_t^f} \min 1, \frac{\pi(\theta, r)}{\pi(\theta_{t-1}, r_t^0)}.$$

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Now the algorithm only requires an acceptance probability δ as input, and a number of iterations M^{adapt} for the adaptation phase.